

Data Entry and Cleanup Protocol

- **Explanation of Data Levels**

- L0 = level zero
 - Raw version of data after initial entry
- L1 = level one
 - Checked and corrected to ensure that it matches field data sheets
- L2 = level two
 - Cleaned and corrected data
 - Changes made to clarify unclear entries, fix obvious mistakes, and replace shorthand (e.g. dashes)
 - Changes made between L1 and L2 are logged in change log

- **Entering the Data: L0**

- Create a Google sheet by making a copy of the [Forest Inventory+Mortality Data Entry Template 2024-05-23](#). This template has one sheet for entering data, one for logging changes, and one for documenting issues.
 - The Google sheet should be named in the following format:
 - ER-APL1_inventory_data_2023_L0_23-12-16
 - ER-APL1 = plot id
 - 2023 = year of data **collection**
 - L0 = level zero, referring to raw version of data that matches the field data sheets
 - 23-12-16 = date of file creation
 - Using format year-month-day
 - The files you create should be stored in the site specific folder on the shared drive.
 - The site-specific folders are located in the [Inventory Plots](#) folder.
 - Copy in the entries from the prior census for the following columns:
Site_Name, Tag_Number, Previous_Tag_Number, Tag_Date, and Sp_Code
 - Family, Genus, Species, etc. will be entered in a subsequent version of the data sheet, using code that is pasted into each column and ensures that these fields match the species code for each individual.
- Enter all data exactly as it is written in the field data sheets.

- If a field is blank on the data sheet leave it blank, but highlight the cell.
- If a dash is entered, enter a dash.
- If a check mark is entered, paste =UNICHAR(10004) into the cell.
- If an entry is unclear, highlight the cell and make a note in the [issue log](#).
- If an entry seems wrong, highlight the cell and make a note in the [issue log](#).
- If a note in the Death_Dam_Mode or Comments field from the prior census is circled, consider it written in and add the note to the Google sheet.
- For data collection date columns (Height_Date, DBH_Date, CII_Date, Canopy_Position_Date, Condition_Obs_Date) enter the date at the top of the field data sheet
 - Unless a different date is noted next to an observation or in the comments field (in that case enter the date noted)
 - If no date is written at the top of the data sheet enter the last day of the census for that site (the latest date noted on any of the data sheets)

- **Issue Log: L0,L1, L2**

- The issue log should be a separate sheet on the same file (see [Data Entry Template](#)).
- There is a column for issues, a column for resolutions, and a column to indicate the date the resolution is implemented.
- Log issues while entering, checking, and cleaning the data.
- Discuss proposed resolutions with a supervisor (e.g. Lara or Tom) before implementation.
- Examples of issues:
 - Data for a specific tree is entered into two separate rows (sometimes one row is entered into the row designated for that tag number and one row is entered into a blank row).
 - The height and DBH for a tree do not seem physiologically possible together. Only note when the combination seems obviously wrong.
 - e.g.) a DBH of 5cm and a height of 30m
 - Entries that are not legible – note in issue log and reach out to field crew for assistance.
 - Wounded trunk is entered by two different teams and they wrote different values
 - in this case enter both values (e.g. 0/1) in L0 and L1 and then change to the higher value in L2

- **Checking the Data: L0 → L1**

- Hide all columns not present on the field datasheets before printing.
 - Print one-sided with narrow margins.
- Check each entry against the field data sheets, moving row by row.
- Keep the L0 Google sheet open while checking data and make any necessary changes if mistakes are identified.
- After the data is checked and corrected to match the field datasheets the level in the name should be changed from L0 to L1

- **Cleaning the Data: L1 → L2**

- Create a new Google sheet, by making a copy of the first file and changing the date in the file name.
 - The Google sheet should be named in the following format:
 - ER-APL1_inventory_data_2023_L2_23-12-16
 - ER-APL1 = plot id
 - 2023 = year of data **collection**
 - L2 = level two, referring to clean data, with changes logged in change log
 - 23-12-16 = date of file creation
 - Using format year-month-day
- If a tree has a comment saying it was retagged and has no entries: keep tree row in the datasheet with the tree tag number and comment saying that it was retagged (with new tag number).
 - Enter NA in all other columns (including species)
- Use the changelog template to record changes made.
 - The changes expected for all plots are listed in the [Changelog Template](#) (a separate sheet in the data entry file).
 - This sheet should be present in your new file if you originally entered data into a copy of [Forest Inventory+Mortality Data Entry Template 2024-05-23](#).
 - Entries made using equations (e.g. DBH_Avg_Ht) should be copied and pasted special (values only) to remove equations
 - Common changes include things like:
 - Replacing blank entries with NA (when appropriate)
 - Rounding entries to the tenths place
 - Entering averages into Height_Avg_M and DBH_Avg_CM fields

○ DBH_Avg_CM calculated using following equation:

$$=IFERROR(IFS(N2="NA","NA",O2="NA",N2,AND(NOT(N2="NA"),NOT(O2="NA")),2*SQRT(((N2/2)^2+(O2/2)^2/2)), "NA")$$

- Replace commas with semicolons
- Changes the wording of a comment or death dam mode entry to increase clarity (e.g. replacing “10% red” with “10% red needles”)
- Notes of ‘dead top’ should be in Comments not DeathDam_Mode.
- Movement of entries that appear to be written into the wrong column on the field data sheet (e.g. entry of S in Survival_Status)
- Resolutions for contradictory entries in the field data sheets:
 - When DeathDam_Status is marked leaning on one field data sheet and standing on another field data sheet enter leaning when entering data to a Google sheet
 - For all duplicate trunk wound entries enter the higher value
- When documenting and implementing changes in the changelog:
 - Enter the date of implementation of the change.
 - Enter the source sheet and the destination sheet. The source sheet is the first file where the data was entered (the copy of the data entry template) and the destination sheet is the new file (the copy of the first file).
 - Enter the cell(s) or column(s) where the change(s) were made.
 - Describe the reason(s) for the change(s) in the comments field.
- After making the changes indicated in the [Changelog Template](#), address each remaining issue (and each highlighted cell) and log changes made individually.
 - All issues not listed in the changelog template should be discussed with a supervisor (e.g. Lara or Tom) to determine resolutions.
 - The tag number of the tree in question should be listed in the comments field.
 - e.g.) 6278 has entry of A in Survival Status and OK in Death Dam Status; switched OK value to Survival Status and inferred standing for Death Dam Status
 - Comments that are unclear can be revised to increase clarity.